

$e \cdot E_0 = e \cdot \nabla \phi$ — условие электроостатности

$$\vec{E}_0 = \vec{E}_B + \vec{E}_0, \quad \vec{j} = \sigma \vec{E}_0 = e n \vec{v}; \quad (j = \frac{dI}{ds})$$

ток. j. Ана. проводимость

Полное поле. Потенциал. поле вклев-во вуден

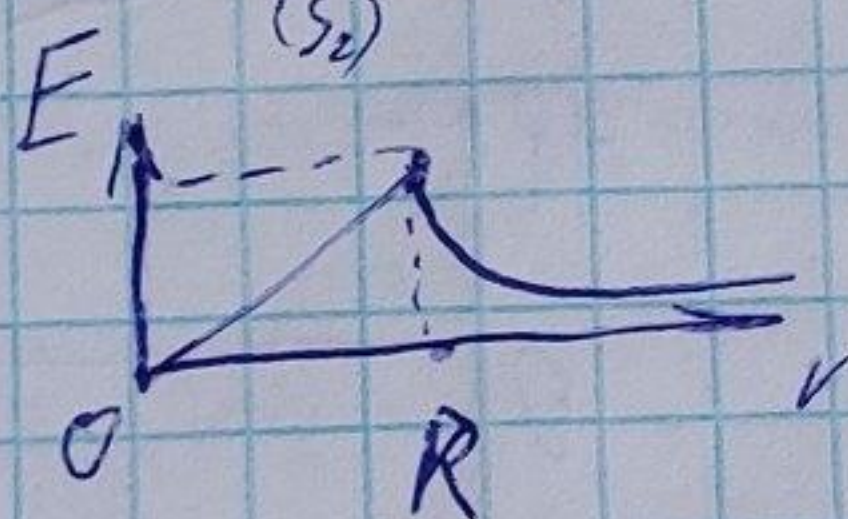
2 режима — при R_L, R_C .

$$\vec{E} = \vec{E}_B + \vec{E}_0$$

$$\vec{E}_{(r_2)} \cdot d\vec{S}_2 = \vec{E}_{(r_2)} \cdot \vec{n}_2 \cdot dS_2 = E_{(r_2)} \cdot n_2 \cdot \cos \alpha \cdot dS = E_{(r_2)} dS$$

$$\oint_{(S_2)} \vec{E}_{(r_2)} \cdot d\vec{S}_2 = \oint_{(S_2)} E_{(r_2)} dS = E_{(r_2)} \oint_{(S_2)} dS = E_{(r_2)} \cdot S_2 = E_{(r_2)} \cdot 4\pi r_2^2$$

$$4\pi r_2^2 E_{(r_2)} = \frac{Q}{\epsilon_0}$$



$$\int_{\psi(\infty)}^{\psi(r_2)} d\psi = - \int_{\infty}^{r_2} E(r) dr;$$

$$\psi \Big|_{\psi(\infty)}^{\psi(r_2)} = - \int_{\infty}^{r_2} \frac{Q}{4\pi r^2 \epsilon_0} dr$$

$$\psi(r_2) = - \frac{Q}{4\pi r^2 \epsilon_0} \int_{\infty}^{r_2} \frac{dr}{r^2};$$